

DTOF RGB-D-IMU CAMERA · S SERIES

# LONG-RANGE MRDVS S10 ULTRA RGB-D CAMERA

Forty-two metres of solid-state depth with a 200 Hz IMU built in. RGB-D-IMU fusion that feeds FAST-LIO and FAST-LIVO directly – rated IP67 and -20 °C to 75 °C for outdoor duty.



## ■ WHY THIS CAMERA

- **0.2–42 m at 90% reflectivity**, 0.2–30 m at 5% – by far the longest range in the S Series, for large-area mapping
- **Built-in IMU at 200 Hz** with RGB-D-IMU fusion – supports FAST-LIO and LIVO (LiDAR-inertial-visual odometry)
- **Fully solid-state design** – no moving parts, for vibration resistance on mowers, drones and rough-terrain vehicles
- **120° × 80° depth FOV, 120° × 110° RGB FOV** – wide colour coverage for semantic recognition alongside geometry
- **IP67 and -20 °C to 75 °C** – the only sealed, high-temperature-rated camera in the S Series
- **≤4 cm accuracy** with 100 kLUX ambient light immunity – stable under direct sunlight at full range
- **Ethernet with PTP and software trigger**; a MIPI CSI-2 module is available for volume programmes on Jetson, Qualcomm, NXP and Rockchip hosts
- **<9 W average draw** at 106 × 69 × 43 mm and 440 g, with a C / C++ / ROS 1 / ROS 2 SDK

**42 m**

MAX. RANGE

**200 Hz**

BUILT-IN IMU

**IP67**

SEALING

**75 °C**

MAX. OPERATING

## ■ APPLICATION SCENARIOS



### ROBOTIC MOWERS

Large-area lawn mapping and navigation



### UAV PLATFORMS

Flight-path planning and terrain following



### OUTDOOR SLAM

FAST-LIO / FAST-LIVO with RGB-D-IMU fusion



### LOW-SPEED VEHICLES

Long-range outdoor collision prevention

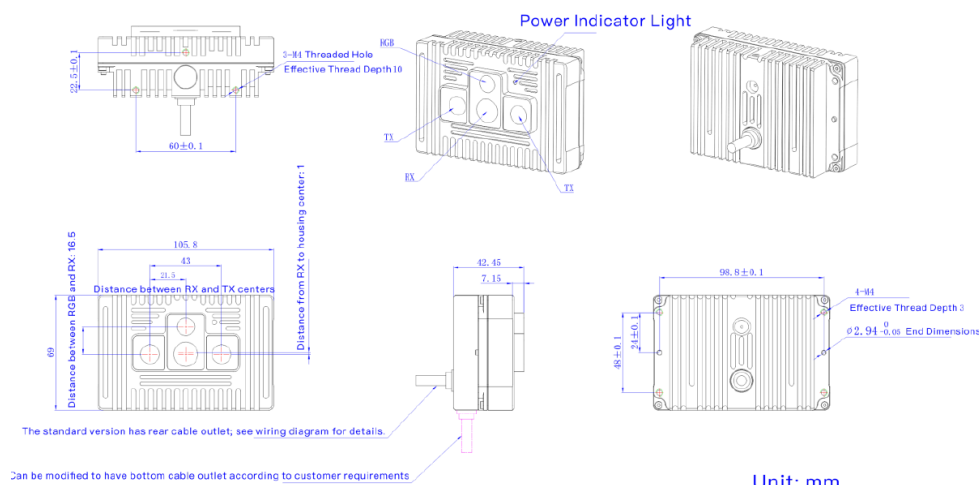
# MRDVS S10 ULTRA — SPECIFICATIONS



| PARAMETER                                  | TECHNICAL DATA                                                    |
|--------------------------------------------|-------------------------------------------------------------------|
| Sensing technology                         | <b>dToF (direct time-of-flight) SPAD array, fully solid state</b> |
| Laser wavelength                           | <b>940 nm</b>                                                     |
| Laser safety level                         | <b>Class 1 (IEC 60825-1:2014)*</b>                                |
| Output format                              | <b>Depth / RGB / IR amplitude map / point cloud</b>               |
| Depth resolution & frame rate <sup>①</sup> | <b>240 × 160 @ max. 10 fps (typ. 10 fps)</b>                      |
| ToF FOV (H × V)                            | <b>120° × 80°</b>                                                 |
| RGB resolution & frame rate <sup>①</sup>   | <b>1280 × 1080 @ max. 10 fps (typ. 10 fps)</b>                    |
| RGB FOV (H × V)                            | <b>120° × 110°</b>                                                |
| Range                                      | <b>0.2–42 m (90% reflectivity)<br/>0.2–30 m (5% reflectivity)</b> |
| Accuracy <sup>②</sup>                      | <b>±4 cm</b>                                                      |
| Ambient light resistance                   | <b>100 kLUX</b>                                                   |
| IMU                                        | <b>Built-in, 200 Hz output; RGB-D-IMU fusion</b>                  |
| Interface                                  | <b>Ethernet (software trigger supported)</b>                      |

| PARAMETER                 | TECHNICAL DATA                             |
|---------------------------|--------------------------------------------|
| MIPI CSI-2 available      | <b>Yes (S10 Ultra MIPI module)</b>         |
| Time synchronisation      | <b>PTP</b>                                 |
| Power supply              | <b>12–27 V DC</b>                          |
| Average power consumption | <b>&lt;9 W</b>                             |
| Dimensions (L × W × H)    | <b>106 × 69 × 43 mm</b>                    |
| Weight                    | <b>440 g</b>                               |
| IP rating                 | <b>IP67</b>                                |
| Operating temperature     | <b>–20 °C to 75 °C</b>                     |
| Storage temperature       | <b>–40 °C to 85 °C</b>                     |
| Humidity                  | <b>Max. 90% RH, operating and storage</b>  |
| SDK support               | <b>C / C++ / ROS 1 / ROS 2</b>             |
| OS compatibility          | <b>Windows 7/8/10/11, Linux, Arm Linux</b> |
| GUI software              | <b>LxCameraViewer</b>                      |

## OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

**Certification.** CE, Class 1 laser safety (IEC 60825-1:2014), RoHS (2011/65/EU Annex II and amendment (EU) 2015/863) and REACH (EC No 1907/2006) certification is in progress, with completion planned for Q3 2026. \* Marked values are design targets pending completion of certification. Contact MorpheusTEK to confirm current certification status.

**Notes.** ① Typical (typ.) frame rate is the recommended operating condition for optimal thermal and power performance; maximum (max.) frame rate is technically achievable but increases power draw and temperature rise and is not recommended for general use. Contact MorpheusTEK for application-specific evaluation. ② Accuracy may degrade with distance and depends on object reflectivity and surface roughness.

**Installation.** Mount the camera centred in its opening with no housing edge, rib or bracket intruding into the nominal field of view; FOV truncation causes fixed blind areas and false close-range readings. Keep the sensor at least 10 cm from highly reflective surfaces. Apply moderate, even mounting torque. Provide ventilation in enclosed spaces. Clean the optical window on a regular schedule in dusty or humid environments.

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